

Student Academic Performance Prediction

RAYBYTE Machine Learning Project • Academic Year 2025–26

GitHub: github.com/heysanzu/RAYBYTE

Project Overview & Educational Impact

RAYBYTE is an advanced Machine Learning initiative designed to bridge the gap between data science and education. By analyzing multi-dimensional student profiles—spanning demographics, classroom behavior, engagement metrics, and attendance records—the project builds robust classification models to predict academic performance classes (Low, Medium, High).

Why it matters: Early prediction of academic standing enables educators and mentors to intervene proactively, offering personalized support to struggling students before academic gaps widen.

Core Objectives

- **Pattern Discovery:** Uncover hidden correlations between daily student engagement (e.g., raised hands, visited resources) and final grades.
- **Algorithm Benchmarking:** Implement and compare 5 distinct classifiers ranging from linear models to neural network architectures.
- **Interactive Predictions:** Provide real-time classification for manually entered student parameters.
- **Data Visualization:** Deliver comprehensive graphical insights including semester-wise, gender-wise, and correlation heatmaps.

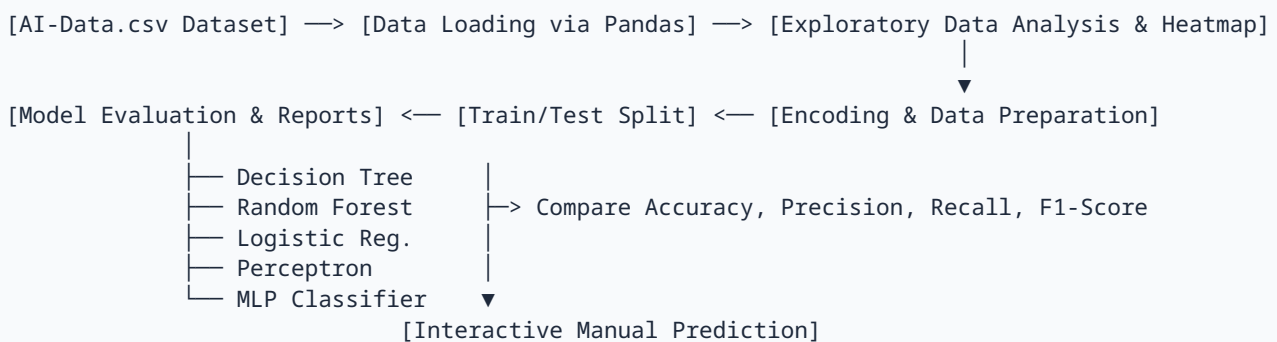
Dataset Structure & Feature Engineering

Category	Key Attributes	Significance
Demographic	Gender, Nationality, Place of Birth	Socio-cultural context and background analysis
Academic	Stage, Grade, Section, Semester	Educational level tracking
Engagement	Raised Hands, Visited Resources, Discussions	Active classroom participation indicators
Attendance	Student Absence Days	Critical factor impacting overall achievement

Machine Learning Models & Architecture

- **Decision Tree Classifier:** Intuitive tree-based rules capturing non-linear feature splits.
- **Random Forest Classifier:** Ensemble learning method combining multiple trees to reduce overfitting and boost predictive stability.
- **Perceptron & Logistic Regression:** Efficient linear classifiers serving as powerful performance baselines.
- **MLP Classifier:** Multi-Layer Perceptron neural network leveraging logistic activation for complex pattern recognition.

Project Workflow



Evaluation Metrics & Understanding

To ensure rigorous model assessment, the project evaluates classifiers using standard metrics:

- **Accuracy:** Overall proportion of correct predictions across all classes.
- **Precision vs. Recall:** Precision ensures reliability of positive class predictions, while recall measures the model's ability to capture all actual instances of a performance tier.
- **F1-Score:** The harmonic mean balancing precision and recall for skewed performance classes.

Installation & Quick Start

```
git clone https://github.com/heysanzu/RAYBYTE.git cd RAYBYTE pip install -r requirements.txt python Project.py
```

Project Team (2025–26)

Role	Member	GitHub Handle
Leader	Shahnewaz	@heysanzu
Co-Leader	Abhishek	@Aprav-X
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